

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 13: Connecting the Dots

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	13 — Connecting the Dots
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Compare per-item means not raw totals when group sizes differ, always sort before taking a median, and keep mean, median and mode distinct — an outlier shifts the mean far more than the median. Do not write on this paper — mark answers on your answer sheet.

1. A student's six test scores have a mean of 72. After a seventh test the mean rises to 74. What did the student score in the seventh test?

- (A) 74 (B) 76
(C) 86 (D) 2

2. The mean age of a family of 5 is 28 years. The mean age of the same family 4 years later (everyone alive) will be:

- (A) 28 years (B) 48 years
(C) 32 years (D) 30 years

3. A list of seven numbers is 4, 4, 6, 7, 9, 9, 9. After one more 4 is added, what is the mode of the eight numbers?

- (A) 4 only (B) 9 only
(C) There are now two modes, 4 and 9 (each appears three times)
(D) 7, the middle value

4. A data set is 3, 5, 6, 8, 10. The value 10 is replaced by 50. How do the mean and the median change?

- (A) Both the mean and the median increase by 8
(B) The mean increases by 8 but the median stays at 6

- (C) The median increases but the mean stays the same
- (D) Neither changes because only one value moved

5. The mean of 12 numbers is 9. The mean of the first 7 of them is 8 and the mean of the last 4 is 11. What is the remaining (8th when counted) number — the one not in either group?

- (A) 9
- (B) 12
- (C) 20
- (D) 8

6. The mean of five numbers is 18. Four of them are 12, 25, 19 and 14. The trap-free fifth number is:

- (A) 18
- (B) 70
- (C) 20
- (D) 90

7. A cricketer's five scores have a range of 64. His lowest score is 6 and his highest is the only other extreme. If his lowest is 6, his highest score is:

- (A) 70
- (B) 64
- (C) 58
- (D) 6

8. Eight observations in increasing order are 5, 8, 11, p , q , 20, 23, 27. Their median is 15. Which is a possible pair (p , q)?

- (A) $p = 15$, $q = 15$ only
- (B) $p = 14$, $q = 16$
- (C) $p = 13$, $q = 18$ with median 15.5
- (D) any p and q because the median uses the 4th value alone

9. The mean weight of 9 boxes is 40 kg. One box weighing 60 kg is taken away. The mean weight of the remaining 8 boxes is:

- (A) 37.5 kg
- (B) 40 kg
- (C) 20 kg
- (D) 45 kg

10. Class A has 25 students with mean score 70; Class B has 15 students with mean score 78. The combined mean score of all 40 students is:

- (A) 74
- (B) 148
- (C) 76
- (D) 73

11. The data 6, 9, 9, 12, 12, 12, 12, 15, 15 has mode 12 (it appears four times). If two more 15s are added, the new mode is:

- (A) 12 only
- (B) Both 12 and 15 (each now appears four times)
- (C) 15 only
- (D) 13.5, the new mean

12. A data set of 6 values is 2, 4, 6, 8, 10, 12. If the largest value 12 is replaced by 120, by how much does the mean increase compared with the median's change?

- (A) Both increase by 18
- (B) The mean increases by 108 and the median by 54

- (C) The mean increases by 18 while the median does not change
(D) Neither changes

13. The mean of 0, 6, 6, 8, x is 6. The value of x is:

- (A) 6 (B) 30
(C) 20 (D) 10

14. The mean of n numbers is 15, and their total is 135. The value of n is:

- (A) 9 (B) 15
(C) 135 (D) 120

15. A double bar graph shows runs by two batters across three matches. Batter X: 40, 50, 30. Batter Y: 20, 70, 30. Which statement is correct?

- (A) X scored more overall because X won two of the three matches
(B) Their totals are equal (120 each) but Y is less consistent because Y's range (50) exceeds X's range (20)
(C) Y is more consistent because of the single 70
(D) Both have equal range so consistency is identical

16. Seven numbers in increasing order are 3, 6, 9, m , 15, 18, 21 and their median is 12. The value of m is:

- (A) 9 (B) 12
(C) 15 (D) 11

17. The mean of 10 numbers was calculated as 50, but one value was wrongly entered as 25 instead of the true 75. The corrected mean is:

- (A) 55 (B) 50
(C) 60 (D) 52

18. Two sections of 30 students each have the same mean mark of 65 but Section P has range 12 and Section Q has range 48. The correct conclusion is:

- (A) Section Q's marks are more consistent
(B) Section P must have a higher total than Section Q
(C) The marks in the two sections are identical
(D) Section P's marks are more consistent (clustered closer to the mean) than Section Q's

19. A bar graph of books read in four months shows Jan 12, Feb 18, Mar 15, Apr 15. Which value is simultaneously the mode and the median of these four numbers?

- (A) 18, because it is the tallest bar (B) 15
(C) 16.5, the mean (D) 12, the shortest bar

20. Group A of 4 numbers has mean 10; group B of 6 numbers has mean 20. The mean of all 10 numbers is:

- (A) 15
(C) 200
- (B) 30
(D) 16

21. A survey of pets per home gives 0, 1, 1, 2, 1, 3, 0, 1, 2, 1. Which statement about the mode and median is correct?

- (A) Mode = 1 and median = 1
(C) Mode = 0 and median = 1
- (B) Mode = 1 and median = 1.5
(D) Mode = 3 because it is the largest value

22. The mean of 6 numbers is 25. When a 7th number is added the mean becomes 27. The 7th number is:

- (A) 27
(C) 2
- (B) 29
(D) 39

23. A bar graph (scale: 1 unit = 5 litres) shows a tank's bar reaching 7.4 units. The volume of water is:

- (A) 37 litres
(C) 12.4 litres
- (B) 7.4 litres
(D) 74 litres

24. A list of five distinct numbers has median 11. If the two smallest are each reduced by 3 and the two largest are each increased by 3, the new median is:

- (A) 11 (the middle value is unchanged)
(C) 14
- (B) 8
(D) 11.5

25. Five numbers have mean 14. A sixth number equal to 14 is added. What happens to the mean?

- (A) It rises to 16.8
(C) It stays exactly 14
- (B) It falls to 11.7
(D) It rises by 14 to 28

26. A magazine bar graph starts its vertical axis at 80. Brand A's bar (value 84) appears three times as tall as Brand B's bar (value 82). The honest reading is:

- (A) Brand A sold three times as much as Brand B
(B) Brand B sold nothing
(C) A and B sold almost the same; the truncated axis exaggerated a tiny 2-unit gap
(D) The graph is impossible to read

27. The mean of a, b and c is 20, and the mean of c, d and e is 30. The mean of all five distinct numbers a, b, c, d, e cannot be found unless we know:

- (A) the value of a only
(B) the value of c, which is counted in both groups
(C) nothing more; it is simply 25
(D) the value of d only

28. Six numbers have mean 30. The five largest of them have mean 33. The smallest number is:

- (A) 30 (B) 33
(C) 3 (D) 15

29. Nine numbers in increasing order have median 40. If the four largest are each increased by 100, the median becomes:

- (A) 140 (B) 65
(C) 40 (unchanged) (D) 60

30. A shop's daily sales (in items) over six days are 20, 35, 0, 35, 30, 50. The owner says the 'typical' day is 35. Which representative value supports this best, and is it the only one equal to 35?

- (A) The mode is 35; the mean (28.3) and median (32.5) are not 35
(B) The mean is 35, matching his claim (C) The median is 35
(D) All three (mean, median, mode) equal 35

31. The mean of five numbers is 16. If 4 is subtracted from each number, the new mean is:

- (A) 16 (B) 12
(C) 80 (D) 60

32. A double bar graph compares boys and girls choosing two activities. Dance: boys 8, girls 22. Sports: boys 24, girls 12. For which activity is the boys-to-girls difference larger, and what is that difference?

- (A) Dance, with a difference of 14 (B) Sports, with a difference of 36
(C) Dance, with a difference of 30 (D) Sports, with a difference of 12

33. The mean of seven numbers is 10. Six of them are 7, 9, 11, 13, 8 and 12. The seventh number is:

- (A) 10 (B) 60
(C) 70 (D) 12

34. Data: 4, 4, 5, 6, 100. Which ordering of the three averages is correct?

- (A) mean < median < mode (B) mode < median < mean
(C) mode = median = mean (D) median < mode < mean

35. The mean of 8 numbers is 16. Three more numbers are added and the mean of all 11 becomes 18. The mean of the three added numbers is:

- (A) 18, the new mean (B) 2, the change in mean
(C) about 23.3 (a total of 70 shared by 3) (D) 70, their total

36. Reading a bar graph (scale 1 cm = 8 marks), four students' bars measure 5 cm, 6 cm, 4 cm and 5 cm. The mean mark is:

- (A) 5 marks (B) 20 marks
(C) 160 marks (D) 40 marks

- 37.** A data set of 5 numbers has mean 20 and range 30, with largest value 38. The smallest value is:
- (A) 20 (B) 30
(C) 68 (D) 8
- 38.** In a class quiz, marks out of 10 are 6, 7, 8, 8, 8, 9, 10, 8. The difference (mean - median) is:
- (A) 1 (B) 0.5
(C) 8 (D) 0
- 39.** The mean of three numbers is 12 and the mean of seven other numbers is 22. The mean of all ten numbers is:
- (A) 19 (B) 17
(C) 34 (D) 190
- 40.** A pictograph uses one symbol = 6 students. Class P shows $4\frac{1}{2}$ symbols and Class Q shows 3 symbols. How many more students are in Class P than in Class Q?
- (A) 9 (B) 1.5
(C) 6 (D) 45
- 41.** The mean of 10 numbers is 12. If one number is changed from 7 to 27, the new mean is:
- (A) 12 (B) 14
(C) 32 (D) 13
- 42.** A data set in increasing order is 8, 10, 12, 14, 16, 18. If the value 8 is removed, what happens to the median?
- (A) It stays at 13 (B) It falls to 12
(C) It rises from 13 to 14 (D) It rises to 15
- 43.** Five children's heights (cm) are 130, 135, 140, 145 and one unknown. The mean is 138. The unknown height is:
- (A) 138 cm (B) 140 cm
(C) 550 cm (D) 690 cm
- 44.** A list of seven numbers has mean 20 and median 18. If the largest number (originally 40) is increased to 60, which statement is true?
- (A) Both the mean and the median increase
(B) The mean increases but the median stays 18
(C) The median increases but the mean stays the same
(D) Neither changes
- 45.** The mode of a list of 11 numbers is 5 and it appears 4 times. If three more 8s are added (8 was not previously the mode), what is the largest possible new frequency of the mode, assuming 8 originally appeared once?

- (A) 8 becomes the unique mode with frequency 3
- (B) 5 stays the unique mode with frequency 4
- (C) There is no mode any more
- (D) The mode is still 5 with frequency 4, since 8 now appears only $1 + 3 = 4$ times — a tie, so both 5 and 8 are modes

46. Two bar graphs for monthly rainfall use different scales: graph 1 has $1 \text{ cm} = 10 \text{ mm}$, graph 2 has $1 \text{ cm} = 25 \text{ mm}$. A 6 cm bar on graph 1 and a 3 cm bar on graph 2 represent rainfalls of:

- (A) 6 mm and 3 mm
- (B) 60 mm and 75 mm respectively
- (C) 60 mm and 30 mm
- (D) the same amount, since 6 cm doubles 3 cm

47. A cricketer's scores over 6 innings are 30, 0, 45, 30, x, 30 with a mode of 30 and a mean of 30. The value of x is:

- (A) 45
- (B) 30
- (C) 180
- (D) 0

48. In a data set, the mean equals 50 and the median equals 50, but the mode equals 30. What can we reasonably conclude?

- (A) The data must be wrong, since all three should be equal
- (B) Most values are 50
- (C) The mode is always between the mean and median
- (D) The value 30 occurs most often, yet the data is balanced enough that the mean and median still coincide at 50

49. The mean of 4 numbers is 12.5. Adding a fifth number makes the mean a whole number, 13. The fifth number is:

- (A) 13
- (B) 0.5
- (C) 50
- (D) 15

50. A pictograph of trees planted uses one symbol = 8 trees, but half-symbols are allowed. Village P shows 5 symbols, Village Q shows $3\frac{1}{2}$ symbols. The total trees planted by both villages is:

- (A) 8.5
- (B) 40
- (C) 28
- (D) 68

51. The marks of 9 students are 11, 13, 13, 14, 16, 16, 16, 18, 20. The value of (mode - median) is:

- (A) 2
- (B) 3
- (C) 0
- (D) 16

52. Section A (40 students) has mean 55; Section B (10 students) has mean 95. A teacher claims the overall mean is 75 (the average of 55 and 95). Why is the teacher wrong, and what is the true mean?

- (A) The teacher is right; the mean is 75
- (B) The true mean is 50
- (C) Because the sections are different sizes; the true mean is 63
- (D) The true mean is 150

53. A data set 6, 8, 10, 12, 14 has a value added so the median rises to 11 while the count becomes even. The added value must be:

- (A) any number that is at least 12 (e.g. 12 or more)
- (B) exactly 11
- (C) exactly 22
- (D) any number 10 or less

54. The mean of 12, 18, x , 24 and 30 is 21. The value of x is:

- (A) 105
- (B) 21
- (C) 84
- (D) 20

55. A bar graph (scale 1 unit = 4 children) shows 'cycles to school' at 6.5 units. A second bar 'takes bus' is exactly twice as tall. How many children take the bus?

- (A) 13
- (B) 26
- (C) 52
- (D) 104

56. Six numbers have mean 9 and the same six numbers (each squared is not asked) — instead, each number is doubled. The new mean is:

- (A) 9
- (B) 11
- (C) 18
- (D) 54

57. A list of five numbers reads 6, 10, 14, 14, x . The mode is 14 and the mean is 12. The value of x is:

- (A) 14
- (B) 12
- (C) 16
- (D) 60

58. A double bar graph shows a town's tree count for two years. Park P: 2023 $\rightarrow$ 120, 2024 $\rightarrow$ 150. Park Q: 2023 $\rightarrow$ 200, 2024 $\rightarrow$ 180. Which conclusion is correct?

- (A) Both parks gained trees because the bars are tall
- (B) Park P's count rose by 30 while Park Q's fell by 20, even though Q still has more trees overall
- (C) Park P now has more trees than Park Q
- (D) Park Q grew faster than Park P

59. A list of six numbers 4, 7, 9, 9, 11, y has mode 9 and median 9. Which value of y keeps BOTH true?

- (A) $y = 8$, which would make the median 8.5, not 9
- (B) $y = 5$, giving median 8 (the mode 9 survives but the median fails)
- (C) y must be exactly 18 for both to hold
- (D) $y = 9$ — then 9 occurs three times (so the mode is 9) and the third and fourth sorted values are both 9 (so the median is 9)

60. A magazine graph with a vertical axis starting at 100 shows City X at 104 looking far taller than City Y at 101. A reporter writes 'City X is much bigger than City Y.' The mathematically honest statement is:

- (A) The two cities differ by only 3 units; the truncated axis makes a tiny gap look huge
- (B) City X is about four times City Y (C) City Y is essentially zero
- (D) The graph proves City X is much bigger

Solutions — Mathematics (Class 7), Chapter 13: Connecting the Dots — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	A	31	B	41	B	51	C
2	C	12	C	22	D	32	A	42	C	52	C
3	C	13	D	23	A	33	A	43	B	53	A
4	B	14	A	24	A	34	B	44	B	54	B
5	D	15	B	25	C	35	C	45	D	55	C
6	C	16	B	26	C	36	D	46	B	56	C
7	A	17	A	27	B	37	D	47	A	57	C
8	B	18	D	28	D	38	D	48	D	58	B
9	A	19	B	29	C	39	A	49	D	59	D
10	D	20	D	30	A	40	A	50	D	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Six tests at mean 72 give a total of $6 \times 72 = 432$. Seven tests at mean 74 give a total of $7 \times 74 = 518$. The seventh score is $518 - 432 = 86$. Picking the new mean 74 ignores that the seventh score must pull the average up. The gap $74 - 72 = 2$ is only the change in mean, not the score, and $76 = 72 + 4$ also confuses the rise in mean with the actual mark.

2. (C) Every one of the 5 members ages by 4 years, so the total rises by $5 \times 4 = 20$ years and the mean rises by $20 \div 5 = 4$. New mean = $28 + 4 = 32$. The mean does not stay 28 because time passes for all. Adding $4 \times 5 = 20$ to the mean itself gives 48, which wrongly adds the whole extra total to one average. 30 would be only a 2-year increase.

3. (C) Originally 9 appears three times and 4 twice. Adding another 4 makes 4 appear three times too, so 4 and 9 are tied as the most frequent — the data is bimodal. Choosing 4 only ignores that 9 still appears three times; choosing 9 only ignores the new third 4. The mode is about frequency, not the middle position, so 7 (the median) is not the mode.

4. (B) Original mean = $(3+5+6+8+10) \div 5 = 32 \div 5 = 6.4$. New mean = $(3+5+6+8+50) \div 5 = 72 \div 5 = 14.4$, an increase of 8. The sorted lists are 3,5,6,8,10 and 3,5,6,8,50; the middle (3rd) value is 6 in both, so the median is unchanged. The median resists a single extreme value, so it does not jump with the mean, and replacing a value certainly does change the mean.

5. (D) Total of all 12 = $12 \times 9 = 108$. First 7 total = $7 \times 8 = 56$; last 4 total = $4 \times 11 = 44$. Those two groups cover 11 numbers totalling $56 + 44 = 100$, so the one leftover number = $108 - 100 = 8$. Answer 9 just echoes the overall mean. 12 and 20 come from miscounting how many numbers each group covers.

6. (C) Five numbers with mean 18 have a total of $5 \times 18 = 90$. The four known numbers add to $12 + 25 + 19 + 14 = 70$, so the fifth is $90 - 70 = 20$. Picking 18 just copies the mean; 70 is the sum of the four known numbers, not the missing one; 90 is the required total of all five, not a single value.

7. (A) Range = highest - lowest, so highest = range + lowest = $64 + 6 = 70$. Taking 64 alone forgets to add back the lowest value. $58 = 64 - 6$ subtracts instead of adds. 6 is the lowest score, not the highest.

8. (B) With 8 values the median is the average of the 4th and 5th values, p and q, so $(p + q) \div 2 = 15$ means $p + q = 30$. The pair 14 and 16 sums to 30 and keeps order ($11 < 14 < 16 < 20$), so it works. 13 and 18 sum to 31, giving median 15.5, not 15. The median is not forced to be 15, 15; many pairs summing to 30 work. It uses two middle values, not the 4th alone.

9. (A) Total of 9 boxes = $9 \times 40 = 360$ kg. Removing the 60 kg box leaves $360 - 60 = 300$ kg over 8 boxes, so the mean is $300 \div 8 = 37.5$ kg. The mean does not stay 40 because the box removed was above average, pulling the average down. $20 = 60 \div 3$ is a stray division and 45 wrongly raises the mean.

10. (D) Class A total = $25 \times 70 = 1750$; Class B total = $15 \times 78 = 1170$. Combined total = 2920 over 40 students, mean = $2920 \div 40 = 73$. The simple average of 70 and 78 is 74, but that wrongly ignores the larger size of Class A, which weights the result toward 70. Adding the two means gives 148, and 76 over-weights Class B.

11. (B) Initially 12 appears four times and 15 appears twice. Adding two more 15s makes 15 appear four times as well, tying 12 and 15 as the most frequent values, so the data is bimodal. Choosing 12 only ignores the two added 15s; choosing 15 only ignores that 12 still appears four times; the mode is about frequency of values, not the mean.

12. (C) Original mean = $42 \div 6 = 7$. New mean = $(42 - 12 + 120) \div 6 = 150 \div 6 = 25$, an increase of 18. The median of six values averages the 3rd and 4th (6 and 8), giving 7 both before and after, so it is unchanged. The median is unaffected because only the largest value moved; the mean rose by the extra 108 spread over 6 values = 18, not the full 108.

13. (D) Five numbers with mean 6 have total $5 \times 6 = 30$. The four known values add to $0 + 6 + 6 + 8 = 20$, so $x = 30 - 20 = 10$. Choosing 6 just copies the mean; 30 is the required total of all five; 20 is the sum of the four known numbers, not the missing fifth.

14. (A) Mean = total $\div$ count, so count = total $\div$ mean = $135 \div 15 = 9$. Choosing 15 confuses the count with the mean; 135 is the total, not the count; $120 = 135 - 15$ wrongly subtracts.

15. (B) X totals $40+50+30 = 120$; Y totals $20+70+30 = 120$ — equal. X's scores range from 30 to 50, a range of 20; Y's range from 20 to 70, a range of 50, so X is the more consistent (less spread) batter. Winning more matches does not raise the season total here, the 70 is a sign of more spread not less, and the ranges are clearly different.

16. (B) With seven values the median is the 4th value, which is m, so $m = 12$ (and $9 < 12 < 15$ keeps the order). Choosing 9 or 15 takes a neighbour of the middle, not the middle itself; 11 is a plausible-looking but unjustified guess for the 4th position.

17. (A) The recorded total = $10 \times 50 = 500$. The entry was 50 too low ($75 - 25 = 50$), so the true total = $500 + 50 = 550$ and the corrected mean = $550 \div 10 = 55$. The mean cannot stay 50 once a value is corrected upward. 60 adds the full 100 difference wrongly; 52 underestimates the correction (the error of 50 spread over 10 is 5, giving 55, not 2).

18. (D) Equal means with the same student count give equal totals, so neither has a higher total. A smaller range means the values are bunched closer together, so Section P (range 12) is more consistent than Section Q (range 48). Equal means do not imply identical marks — the same average can arise from very different spreads.

19. (B) The value 15 appears twice (Mar and Apr), so it is the mode. Sorting gives 12, 15, 15, 18; the median averages the two middle values (15 and 15) = 15. So 15 is both. The tallest bar (18) is the maximum, not the median or mode; 16.5 is the mean; 12 is the minimum.

20. (D) Group A total = $4 \times 10 = 40$; group B total = $6 \times 20 = 120$. Combined = 160 over 10 numbers = 16. The plain average of 10 and 20 is 15, but B has more numbers so the combined mean leans toward 20. 30 adds the means; 200 is the combined total, not the mean.

21. (A) The value 1 appears five times — more than any other — so the mode is 1. Sorting: 0,0,1,1,1,1,1,2,2,3; with 10 values the median averages the 5th and 6th (both 1) = 1. So both are 1. The median is 1, not 1.5, because the 5th and 6th values are both 1; mode is about frequency, so 0 (two times) and 3 (the largest) are not the mode.

22. (D) Six numbers total $6 \times 25 = 150$; seven numbers total $7 \times 27 = 189$. The 7th number = $189 - 150 = 39$. Picking 27 copies the new mean; 29 wrongly assumes the score equals the mean plus the count of extra; 2 is just the change in mean.

23. (A) Each unit represents 5 litres, so $7.4 \text{ units} = 7.4 \times 5 = 37 \text{ litres}$. Reading the height as the answer (7.4) ignores the scale; $12.4 = 7.4 + 5$ adds the scale instead of multiplying; 74 multiplies by 10 rather than 5.

24. (A) The median of five numbers is the 3rd value when sorted. Only the two smallest and the two largest were altered; the middle value was untouched and the order is preserved, so the median stays 11. Subtracting 3 (giving 8) or adding 3 (giving 14) wrongly changes the middle value, and there is no fractional shift to give 11.5.

25. (C) Adding a value equal to the current mean does not change the mean: total goes from $5 \times 14 = 70$ to $70 + 14 = 84$ over 6 numbers, and $84 \div 6 = 14$. The mean only moves if the added value differs from it. $16.8 = 84 \div 5$ uses the wrong count; the other options misapply the change.

26. (C) Because the axis begins at 80, the visible bar heights measure only the part above 80 (4 units vs 2 units), making A look much taller. The true values 84 and 82 differ by only 2, so the brands sold almost the same. A did not sell three times as much, B clearly sold 82, and the graph can be read once the axis start is noticed.

27. (B) The two groups overlap at c. Sum of first group = 60, sum of second = 90, but adding them counts c twice, so $a+b+c+d+e = 60 + 90 - c = 150 - c$. The five-number mean = $(150 - c) \div 5$, which needs c. It is not simply 25 (that would require no overlap); knowing a or d alone does not remove the double-counted c.

28. (D) Total of six = $6 \times 30 = 180$. Total of the five largest = $5 \times 33 = 165$. The smallest = $180 - 165 = 15$. The smallest is not 30 (the overall mean) nor 33 (a group mean); 3 is just the difference of the two means, not the value removed.

29. (C) With nine values the median is the 5th. Increasing only the four largest (positions 6 to 9) leaves the 5th value untouched and order preserved, so the median stays 40. Adding 100 to the median (giving 140) is wrong because the median value itself was not among those changed; 65 and 60 are unfounded shifts.

30. (A) The value 35 appears twice, more than any other, so the mode is 35. The mean = $(20+35+0+35+30+50) \div 6 = 170 \div 6 \approx 28.3$. Sorting: 0,20,30,35,35,50; median = $(30+35) \div 2 = 32.5$. Only the mode equals 35, so the other choices that claim the mean, median, or all three equal 35 are wrong.

31. (B) Subtracting 4 from each of the 5 numbers lowers the total by $5 \times 4 = 20$, so the mean drops by $20 \div 5 = 4$, giving $16 - 4 = 12$. The mean does change when every value changes. $80 = 5 \times 16$ is the original total; $60 = 80 - 20$ is the new total, not the new mean.

32. (A) Dance difference = $|8 - 22| = 14$; Sports difference = $|24 - 12| = 12$. Dance has the larger gap, 14. The 36 mistakenly adds boys and girls for sports; 30 adds the dance values; 12 is the smaller (sports) gap, not the larger one.

33. (A) Total of seven = $7 \times 10 = 70$. The six known values add to $7+9+11+13+8+12 = 60$, so the seventh = $70 - 60 = 10$. Here the missing value happens to equal the mean. 60 is the sum of the six known numbers; 70 is the required total; 12 is just one of the listed numbers.

34. (B) Mode = 4 (appears twice); sorted 4,4,5,6,100 gives median = 5 (middle value); mean = $119 \div 5 = 23.8$. So $4 < 5 < 23.8$, i.e. mode < median < mean. The large outlier 100 drags the mean far above the others, so the mean cannot be smallest, and the three are clearly not equal.

35. (C) Total of 8 = $8 \times 16 = 128$; total of 11 = $11 \times 18 = 198$. The three added numbers total $198 - 128 = 70$, so their mean = $70 \div 3 \approx 23.3$. The added group must average above 18 to pull the mean up. 18 is the overall new mean, 2 is just the change in mean, and 70 is the added total, not its mean.

36. (D) Convert heights to marks: $5 \times 8 = 40$, $6 \times 8 = 48$, $4 \times 8 = 32$, $5 \times 8 = 40$. Mean of marks = $(40+48+32+40) \div 4 = 160 \div 4 = 40$. Averaging the heights first ($20 \text{ cm} \div 4 = 5 \text{ cm}$) then forgetting to apply the scale gives 5; multiplying that 5 by 4 gives 20; 160 is the total marks, not the mean.

37. (D) Range = largest - smallest, so smallest = largest - range = $38 - 30 = 8$. The smallest is not the mean (20), nor the range (30); $68 = 38 + 30$ adds instead of subtracting.

38. (D) Sorted the marks are 6, 7, 8, 8, 8, 8, 9, 10. Mean = $(6+7+8+8+8+8+9+10) \div 8 = 64 \div 8 = 8$. With 8 values the median averages the 4th and 5th (both 8) = 8. So mean - median = $8 - 8 = 0$. The difference is not a whole 1 or 0.5 (the mean and median coincide here), and 8 is the value of each, not their difference.

39. (A) First group total = $3 \times 12 = 36$; second group total = $7 \times 22 = 154$. Combined = 190 over 10 numbers = 19. The plain average of 12 and 22 is 17, but the seven-number group is larger so the combined mean leans toward 22. 34 adds the means; 190 is the total, not the mean.

40. (A) Class P = $4.5 \times 6 = 27$ students; Class Q = $3 \times 6 = 18$ students; difference = $27 - 18 = 9$. Subtracting the symbol counts ($4.5 - 3 = 1.5$) forgets each symbol is 6 students; 6 is the value of one symbol; 45 is the total of both classes, not the difference.

41. (B) Changing 7 to 27 raises the total by 20, so the mean rises by $20 \div 10 = 2$, giving $12 + 2 = 14$. The mean does change because the total changed. $32 = 12 + 20$ adds the whole difference to the mean; 13 adds only half the correct shift.

42. (C) Original six values: median = $(12+14) \div 2 = 13$. After removing 8, the five values are 10,12,14,16,18 with median = 14 (the middle value). So the median rises from 13 to 14. It does not stay 13 because the count changed from even to odd; it does not fall, since a small value was removed; 15 is not a middle value of either set.

43. (B) Total = $5 \times 138 = 690$ cm. The four known heights add to $130+135+140+145 = 550$, so the unknown = $690 - 550 = 140$ cm. 138 just copies the mean; 550 is the sum of the four known heights; 690 is the required total of all five.

44. (B) Increasing the largest value raises the total, so the mean goes up. The median of seven numbers is the 4th value; the largest value sits at position 7, so moving it further out does not touch the 4th value — the median stays 18. The mean must change because the total changed, and the median is shielded from a single extreme.

45. (D) 5 appears 4 times. 8 appeared once and gains three more, reaching $1 + 3 = 4$. Now 5 and 8 are tied at 4 each, so the data is bimodal — both are modes. 8 alone is not the mode because its count only reaches 4, equalling 5; 5 is not the unique mode because 8 has caught up; and a tie still means there are modes, not none.

46. (B) Graph 1: $6 \times 10 = 60$ mm. Graph 2: $3 \times 25 = 75$ mm. The taller-looking 6 cm bar actually represents less than the 3 cm bar because the scales differ. Reading heights as values gives 6 and 3; using graph 1's scale for both gives 30; and bar height alone is meaningless without matching the scale.

47. (A) Mean 30 over 6 innings needs a total of 180. The five known scores add to $30+0+45+30+30 = 135$, so $x = 180 - 135 = 45$. This keeps 30 as the most frequent value (three times), so the mode condition still holds. 30 would make the total 165, not 180; 180 is the required total; 0 gives a total of 135.

48. (D) Mean, median and mode measure different things, so they need not agree. A frequent low value (mode 30) can coexist with a symmetric balance that keeps the mean and median both at 50. There is no rule forcing all three to be equal, the most common value is 30 not 50, and the mode need not lie between the other two.

49. (D) Four numbers total $4 \times 12.5 = 50$; five numbers total $5 \times 13 = 65$. The fifth number = $65 - 50 = 15$. Picking 13 copies the new mean; 0.5 is just the change in mean; 50 is the four-number total, not the fifth value.

50. (D) Village P = $5 \times 8 = 40$; Village Q = $3.5 \times 8 = 28$; total = $40 + 28 = 68$. Adding the symbol counts ($5 + 3.5 = 8.5$) ignores the scale; 40 and 28 are the separate village totals, not the combined total.

51. (C) The mode is 16 (appears three times). With 9 values the median is the 5th, which is 16. So mode - median = $16 - 16 = 0$. Taking 14 as the median (the 4th value) gives 2; mixing up counts gives 3; 16 is the value of each, not their difference.

52. (C) Averaging 55 and 95 to get 75 ignores that Section A has four times as many students, so the combined mean must lean toward 55. True total = $40 \times 55 + 10 \times 95 = 2200 + 950 = 3150$ over 50 students = 63. So 75 is wrong, 50 is below both means (impossible), and 150 is the sum of the two means.

53. (A) Original median is 10 (middle of five). Adding a 6th value makes the median the average of the 3rd and 4th sorted values. To get median 11, those two must be 10 and 12, which happens when the new value is at least 12 (it slots above 12 or equals it, leaving 10 and 12 in the middle). A value of 11 would itself become a middle value and give median $(10+11)/2 = 10.5$, not 11; 22 works but is only one of many; a value of 10 or less keeps the middles at 8 and 10.

54. (B) Five numbers with mean 21 total $5 \times 21 = 105$. The four known values add to $12+18+24+30 = 84$, so $x = 105 - 84 = 21$. Here 105 is the required total and 84 is the sum of the known four — not x itself; 20 is an unfounded guess.

55. (C) Cycles = $6.5 \times 4 = 26$ children. The bus bar is twice as tall, so 13 units, giving $13 \times 4 = 52$ children (or simply double 26). Doubling the unit height ($6.5 \times 2 = 13$) without applying the scale gives 13; 26 is the cycling count, not the bus count; 104 doubles too far.

56. (C) Doubling every value doubles the total, so the mean also doubles: $9 \times 2 = 18$. The mean does not stay 9 since all values changed; adding 2 (giving 11) confuses doubling with adding; $54 = 6 \times 9$ is the original total, not the new mean.

57. (C) Mean 12 over 5 numbers needs total 60. The four known values add to $6+10+14+14 = 44$, so $x = 60 - 44 = 16$. The mode 14 is already secured by the two 14s, and 16 keeps 14 the most frequent. 14 would make the total 58, not 60; 12 copies the mean; 60 is the required total.

58. (B) Park P went $120 \rightarrow 150$, a rise of 30; Park Q went $200 \rightarrow 180$, a fall of 20. So P grew while Q declined, yet Q's 2024 total (180) still exceeds P's (150). Tall bars do not mean both grew; P does not overtake Q in 2024; and Q shrank, so it did not grow faster.

59. (D) Sort 4, 7, 9, 9, 11 with y inserted. For the median (average of the 3rd and 4th of six values) to be 9, both middle values must be 9, which needs $y \geq 9$. For the mode to stay uniquely 9, y must not create a rival pair — $y = 11$ would tie 11 with 9, and $y = 9$ makes 9 appear three times. The clean value satisfying both is $y = 9$: list becomes 4, 7, 9, 9, 9, 11, mode 9 and median $(9+9)/2 = 9$. With $y = 8$ the middles are 8 and 9, median 8.5; with $y = 5$ the median drops to 8; 'exactly 18' is an arbitrary single value with no special role.

60. (A) Visible bar heights measure only the part above 100 (4 vs 1), so X looks four times Y, but the real values 104 and 101 differ by just 3. The honest reading notes the truncated axis exaggerates a small difference. X is not four times Y, Y is 101 (not near zero), and the graph misleads rather than proves a big gap.